

Calculations for Kryptos K4

When the plaintext is known for a section of code, working out how to decode from

“OBKRUOXOGHULBSOLIFBBWFLRVQQPRNGKSSOTWTQSQSSEKZZWATJKLUDIAWIN
FBNYPVTTMZFPKWGDKZXTJCDIGKUHUAUEKCAR”

We know that the letters at positions 64 – 74 (NYPVTT MZFPK) of this cipher text spell out the words BERLIN CLOCK but so far no public solution has been given to show how this is calculated. This documents my attempt so far to bring clarity to the cipher text as well as organise my own thoughts in order to fill out the missing pieces.

Given that we know the plaintext for a section of the sculpture, we can reverse engineer how the sculpture was encrypted by determining which character, when combined with the related position in the cipher text results in the plaintext character.

Each character in the cipher text has two positions, that from the start of the message and it's related position from the end of the message. Therefore, the N given above is at position 64. The message length is 97 characters resulting in a secondary position of 33 ($97 - 64 = 33$).

Ignoring the secondary position for now, it is worth noting that alongside this, each character in the alphabet has an inverse character. This is important as it leads to the possibility of being able to calculate the X,Y position of each letter in the message when placed on a grid.

For working out this part of the solution, I used a 26x26 Vigenere grid written out on paper.

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26
A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A
2	2	2	2	2	2	2	1	1	1	1	1	1	1	1	1	1	9	8	7	6	5	4	3	2	1
6	5	4	3	2	1	0	9	8	7	6	5	4	3	2	1	0									

Starting with the letter N at position 64, to calculate it's X,Y coordinates, subtract this from its position and use modulus 26 on the resulting value.

For example, N is the 14th character in the alphabet, therefore it's X coordinate is calculated as:

$$X_1 = (64 - 14) = (50 \bmod 26) = 24 = X.$$

Because each character has 2 positions, we flip position 24 (X) to give C and add N to this in order to obtain the Y coordinate. Because we are working with a paper grid, this flip is carried out as $(26 - X_1 \bmod 26) + 1$.
Therefore, $Y_1 = (3 + 14) = (17 \bmod 26) = 17 = Q$.

The coordinates for N are now (24, 17).

Following this through for positions 65 – 74 we get:

	X	Y
64 (N)	$(64 - 14) = (50 \bmod 26) = \mathbf{24} = X$	$(03 + 14) = (17 \bmod 26) = \mathbf{17} = Q$
65 (Y)	$(65 - 25) = (40 \bmod 26) = \mathbf{14} = N$	$(13 + 25) = (38 \bmod 26) = \mathbf{12} = L$
66 (P)	$(66 - 16) = (50 \bmod 26) = \mathbf{24} = X$	$(03 + 16) = (19 \bmod 26) = \mathbf{19} = S$
67 (V)	$(67 - 22) = (45 \bmod 26) = \mathbf{19} = S$	$(08 + 22) = (30 \bmod 26) = \mathbf{04} = D$
68 (T)	$(68 - 20) = (48 \bmod 26) = \mathbf{22} = V$	$(05 + 20) = (25 \bmod 26) = \mathbf{05} = E$
69 (T)	$(69 - 20) = (49 \bmod 26) = \mathbf{23} = W$	$(04 + 20) = (24 \bmod 26) = \mathbf{24} = X$
70 (M)	$(70 - 13) = (57 \bmod 26) = \mathbf{05} = E$	$(22 + 13) = (35 \bmod 26) = \mathbf{09} = I$
71 (Z)	$(71 - 26) = (45 \bmod 26) = \mathbf{19} = S$	$(08 + 26) = (34 \bmod 26) = \mathbf{08} = H$
72 (F)	$(72 - 06) = (66 \bmod 26) = \mathbf{14} = N$	$(13 + 06) = (19 \bmod 26) = \mathbf{19} = S$
73 (P)	$(73 - 16) = (57 \bmod 26) = \mathbf{05} = E$	$(22 + 16) = (38 \bmod 26) = \mathbf{12} = L$
74 (K)	$(74 - 11) = (63 \bmod 26) = \mathbf{11} = K$	$(16 + 11) = (27 \bmod 26) = \mathbf{01} = A$

Finding the square from plaintext.

Because we have part of the plaintext visible, we can figure out how each cipher text character is decoded.

First take the plaintext letter and “square it”

For example, B has the square

X	B
B	F

This square is calculated as:

Top Right:

$$X = ((64 - 2) \bmod 26) = 10.$$

$$Y = ((17 + 2) \bmod 26) = 19.$$

Bottom Left:

$$X = ((64 + 2) \bmod 26) = 14.$$

$$Y = ((13 + 2) \bmod 26) = 15.$$

$$\text{Top Left} = (10, 15) = X$$

$$\text{Bottom Right} = (14, 19) = F$$

This follows through for the remainder of the revealed plaintext letters as such:

x b b f	u e e o	h r r b	n l l j
q i i a	l n n p	w c c i	n l l j
k o o s	w c c i	o k k g	

The character out of this square we are most interested in is the character in the top left position. For the purposes of this calculation we will refer to this as *e*.

The use of the top left position has come from the decoded text to K3 which discusses "I MADE A TINY BREACH IN THE UPPER LEFT HAND CORNER".

Decoding plaintext.

Adding this to the cipher text reveals the plaintext when using the following equation.

$$X_3 = ((c + e) \bmod 26 + p) - c \bmod 26$$

$$X_3 = (14 + 24) = (38 \bmod 26) = (12 + 64) = (76 - 14) = (62 \bmod 26) = 10$$

$$Y_3 = (26 - 10) + 1 = (16 + 14) = (30 \bmod 26) = 5$$

Looking on the grid, this reveals B at position $(X_1, Y_3) = (24, 5)$. Alternatively this can be determined by running $(Y_3 + X_1 \bmod 26)$. Don't forget to remove 1 if doing this programmatically or you'll always end up one character out on Y. For example: $(5 + 24) - 1 = (28 \bmod 26) = 2 = B$.

This ends the primary calculations I have ascertained within the last 120 – 150 hours but does not end the details of the search I am prepared to share.

Other aspects of interest

Etymology

An anagram of the misspellings and their respective counterparts is LAQUEO.

Laqueo (Spanish): First person singular present indicative form of **Laquear**

Laquear (English): (plural laquears) (architecture) A **lacunar**.

Lacunar: (medicine) Of or pertaining to a **lacuna**

Lacuna: (plural lacunae or lacunas)

- A small opening; a small pit or depression; a small blank space; a gap or vacancy; a hiatus.
- **An absent part, especially in a book or other piece of writing, often referring to an ancient manuscript or similar such.**
- (microscopy) A space visible between cells, allowing free passage of light.
- (linguistics) A language gap, which occurs when there is no direct translation in the target language for a lexical term found in the source language

Whilst this may seem to be reaching (and it is certainly convoluted), it struck me as interesting due to the meaning highlighted in bold above which seems to tie in with the etymology of the keywords Abscissa and Palimpsest.

Distances

One of the more speculative areas I have been examining is distances between cipher text, plaintext and the respective encoding letter, which has revealed some seemingly intriguing results.

Consider the following table:

Cipher text	Plaintext	Encoding	Cipher - Encoding	Plain - Cipher	Plain - encoding
N (14)	B (02)	X (24)	14 - 24 = 10 J	02 - 14 = 12 L	02 - 24 = 22 V
Y (25)	E (05)	U (21)	25 - 21 = 22 V	05 - 25 = 20 T	05 - 18 = 13 M
P (16)	R (18)	H (08)	16 - 08 = 18 R	18 - 16 = 24 X	18 - 08 = 16 P
V (22)	L (12)	N (14)	22 - 14 = 18 R	12 - 22 = 10 J	12 - 14 = 02 B
T (20)	I (09)	Q (17)	20 - 17 = 23 W	09 - 20 = 11 K	09 - 17 = 08 H
T (20)	N (14)	L (12)	20 - 12 = 18 R	14 - 20 = 06 F	14 - 12 = 24 X
M (13)	C (03)	W (23)	13 - 23 = 10 J	03 - 13 = 10 J	03 - 23 = 20 T
Z (26)	L (12)	N (14)	26 - 14 = 14 N	12 - 26 = 14 N	12 - 14 = 02 B
F (06)	O (15)	K (11)	06 - 11 = 05 E	15 - 06 = 17 Q	15 - 11 = 22 V
P (16)	C (03)	W (23)	16 - 23 = 07 G	03 - 16 = 13 M	03 - 23 = 20 T
K (11)	K (11)	O (15)	11 - 15 = 04 D	11 - 11 = 26 X	11 - 15 = 04 D

One of the interesting things I have seen within this table is when looking at the distances between the plaintext and cipher text characters, focusing on the two L's and the C's, we can see that although the letters are encoded using the same character, (L encodes with N, C encodes with W), the plaintext to cipher text distances are exactly the number of letters apart.

For example, L at 67 encodes with N whilst the distance from plaintext to cipher text is J. Move forward four characters to position 71 and we hit L again. Here the distance from plaintext to cipher text is N. $J - N = (10 - 14) = 4$.

Similarly we see the same pattern in encoding C at positions 70 and 73 where the plaintext to cipher text distance has moved on exactly 3 places.

Without more of the plaintext to verify this, it would seem to indicate that each use of a letter resets a counter and the next instance of that particular letter would decipher with $c_1 = (c_0 + (p_1 - p_0))$.